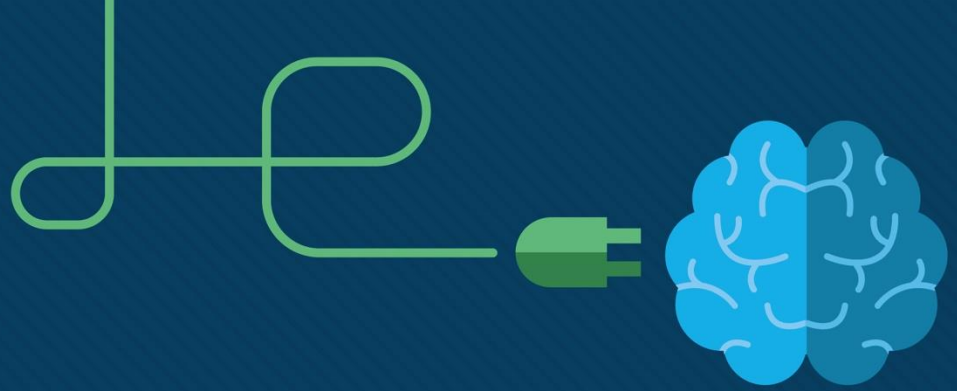
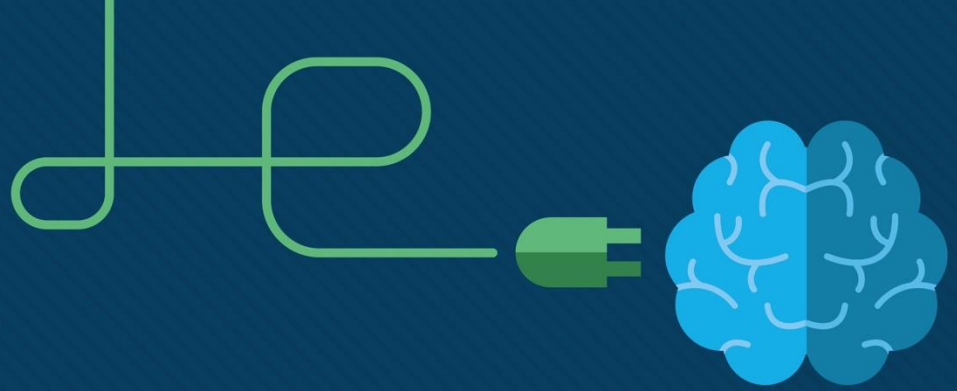


Module 6: Data Link Layer

Instructor Materials



Introduction to Networks v7.0
(ITN)



Module 6: Data Link Layer



Introduction to Networks v7.0
(ITN)

Module Objectives

Module Title: Data Link Layer

Module Objective: Explain how media access control in the data link layer supports communication across networks.

Topic Title	Topic Objective
Purpose of the Data Link Layer	Describe the purpose and function of the data link layer in preparing communication for transmission on specific media.
Topologies	Compare the characteristics of media access control methods on WAN and LAN topologies.
Data Link Frame	Describe the characteristics and functions of the data link frame.

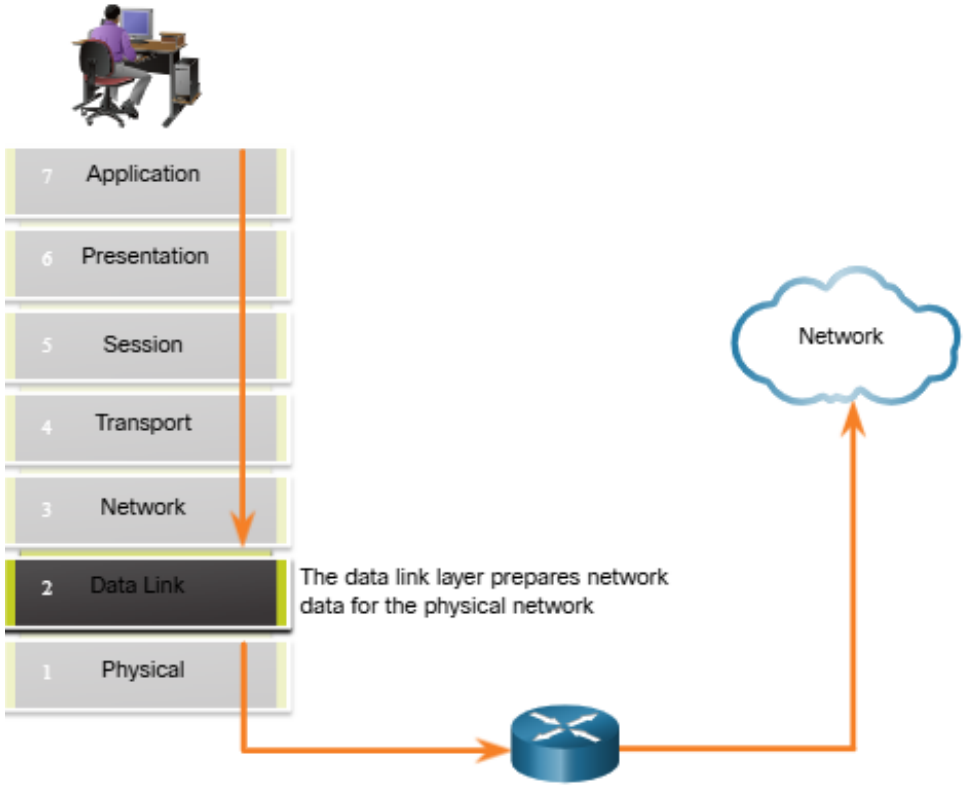
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6.1 Purpose of the Data Link Layer

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The Data Link Layer

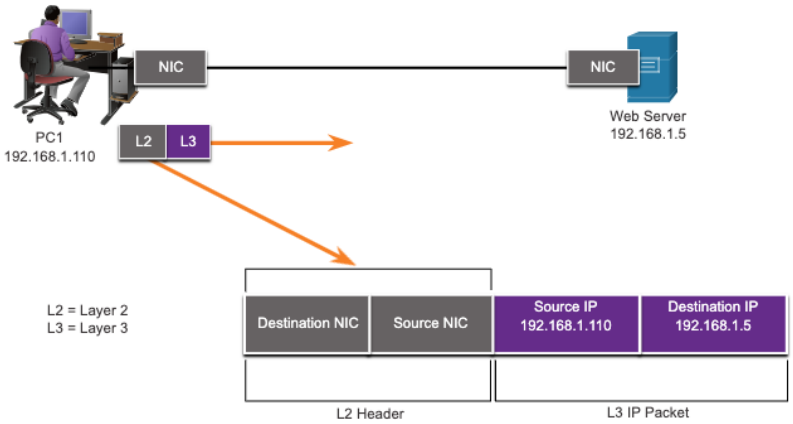
- The Data Link layer is responsible for communications between end-device network interface cards.
- It allows upper layer protocols to access the physical layer media and encapsulates Layer 3 packets (IPv4 and IPv6) into Layer 2 Frames.
- It also performs error detection and rejects corrupts frames.



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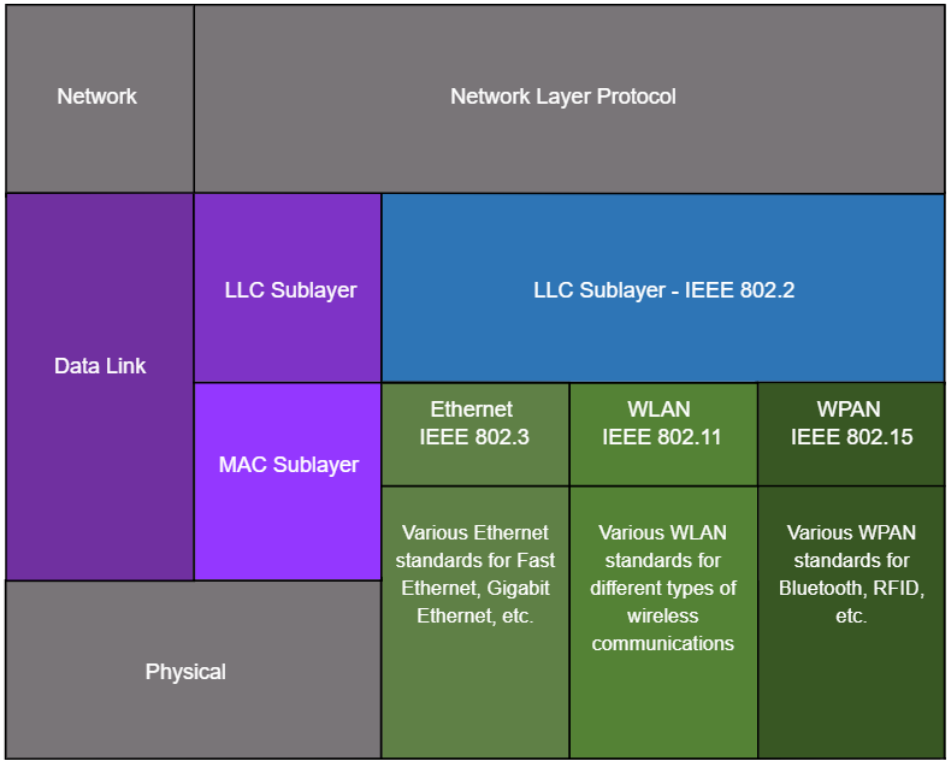
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IEEE 802 LAN/MAN Data Link Sublayers

IEEE 802 LAN/MAN standards are specific to the type of network (Ethernet, WLAN, WPAN, etc).

The Data Link Layer consists of two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC).**

- The LLC sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers.
- The MAC sublayer is responsible for data encapsulation and media access control.



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Functions of the Data Link Layer

1. Framing
2. Link access
3. Flow control
4. Reliable delivery
5. Half-duplex and full-duplex
6. Error detection and correction (rejects corrupt frames)
   - CRC
   - Checksum
   - Parity-checks

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Providing Access to Media

Packets exchanged between nodes may experience numerous data link layers and media transitions.

At each hop along the path, a router performs four basic Layer 2 functions:

- Accepts a frame from the network medium.
- De-encapsulates the frame to expose the encapsulated packet.
- Re-encapsulates the packet into a new frame.
- Forwards the new frame on the medium of the next network segment.

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Data Link Layer Standards

Data link layer protocols are defined by engineering organizations:

- Institute for Electrical and Electronic Engineers (IEEE).
- International Telecommunications Union (ITU).
- International Organizations for Standardization (ISO).
- American National Standards Institute (ANSI).



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6.2 Topologies

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Physical and Logical Topologies

The topology of a network is the arrangement and relationship of the network devices and the interconnections between them.

There are two types of topologies used when describing networks:

- **Physical topology** – shows physical connections and how devices are interconnected.
- **Logical topology** – identifies the virtual connections between devices using device interfaces and IP addressing schemes.

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WAN Topologies

There are three common physical WAN topologies:

- **Point-to-point** – the simplest and most common WAN topology. Consists of a permanent link between two endpoints.
- **Hub and spoke** – similar to a star topology where a central site interconnects branch sites through point-to-point links.
- **Mesh** – provides high availability but requires every end system to be connected to every other end system.

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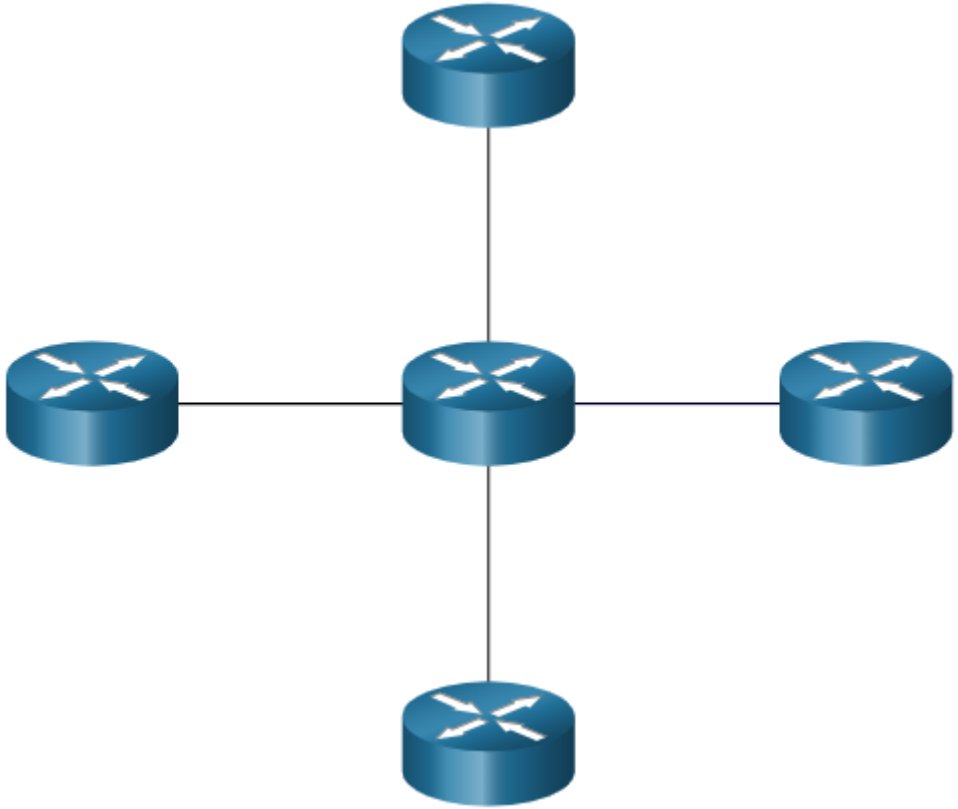
Point-to-Point WAN Topology

- Physical point-to-point topologies directly connect two nodes.
- The nodes may not share the media with other hosts.
- Because all frames on the media can only travel to or from the two nodes, Point-to-Point WAN protocols can be very simple.



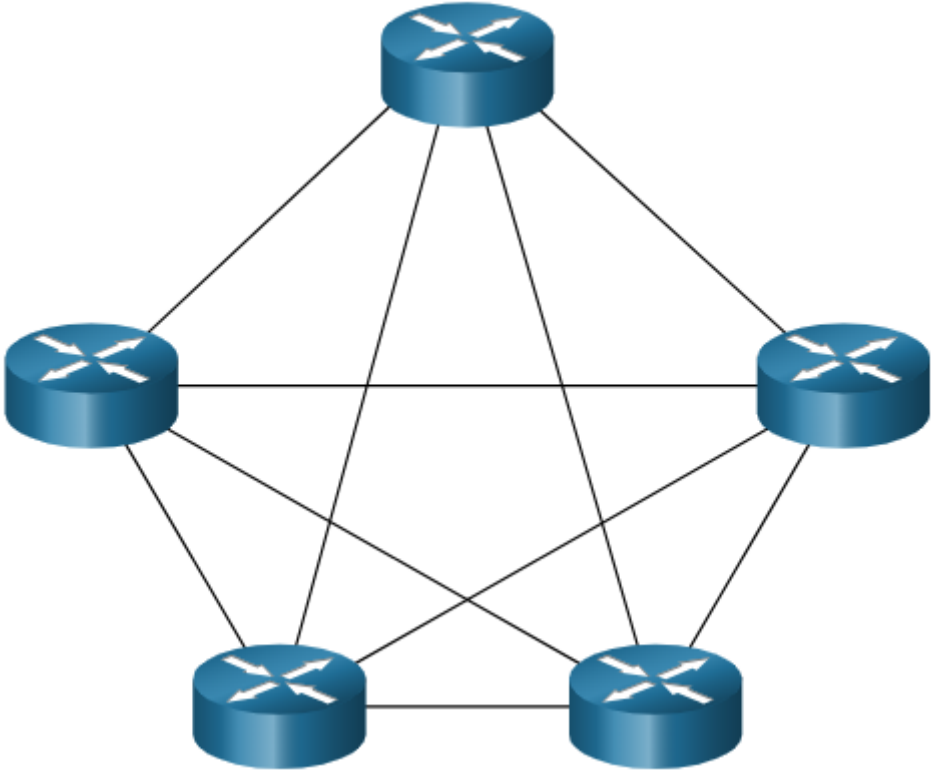
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Hub and Spoke WAN Topology



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Hub and Spoke WAN Topology



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LAN Topologies

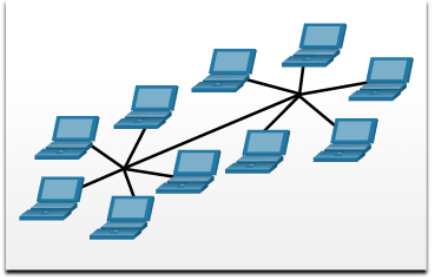
End devices on LANs are typically interconnected using a star or extended star topology. Star and extended star topologies are easy to install, very scalable and easy to troubleshoot.

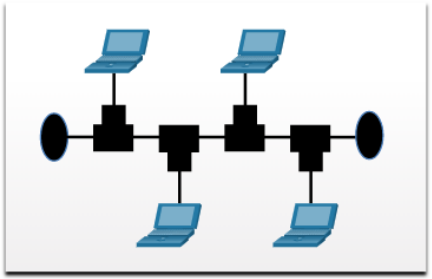
Early Ethernet and Legacy Token Ring technologies provide two additional topologies:

- **Bus** – All end systems chained together and terminated on each end.
- **Ring** – Each end system is connected to its respective neighbors to form a ring.

Physical Topologies


Star Topology


Extended Star Topology


Bus Topology


Ring Topology

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Half and Full Duplex Communication

Half-duplex communication

- Only allows one device to send or receive at a time on a shared medium.
- Used on WLANs and legacy bus topologies with Ethernet hubs.

Full-duplex communication

- Allows both devices to simultaneously transmit and receive on a shared medium.
- Ethernet switches operate in full-duplex mode.

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Multiple Access Problem

Broadcast Channels

- Traditional TV – uses one-way broadcast channel
- Computer networks use two-way multiple broadcast channels (can work as both the sender and reciever)
- Because all nodes can transmit frames, more than two nodes can transmit frames at the same time
  - When this happens, all nodes receive multiple frames at the same time – **collisions**
  - When there is a collision, none of the receiving nodes can make sense of the received frames – are **discarded, a waste of the channel.**

Multiple Access Protocols

- ***Channel partitioning protocols*** – e.g. Time-Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM)
- **Random access protocols** – e.g. Carrier Sense Multiple Access (**CSMA)** protocol, CSMA/CD, CSMA/CA
- ***Taking-turns protocols*** – e.g. Polling protocols, token-passing protocols

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Multiple Access Problem

Two important rules for human conversation

- **Listen before speaking – If someone else speaks, wait until they finished**
  - In networks, this is called **carrier sensing**
  - A node listens to the channel before transmitting
    1. If a frame from another node is currently being transmitted in the channel, the next node waits (or "***backs-off***") a random amount of time and then senses the channel again.
    2. If the channel is sensed to be idle, then the node begins to transmit.
    3. Otherwise, the node waits another random amount of time and repeats this process.
- **If someone else begins speaking at the same time, stop talking**
  - In networks, this is called **collision detection**
  - A transmitting node listens to the channel while it is transmitting
  - If it detects that another node is transmitting at the same time, it stops its transmission and uses some protocol to determine when it should next attempt to transmit

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Access Control Methods

Contention-based access

All nodes operating in half-duplex, competing for use of the medium. Examples are:

- Carrier sense multiple access with collision detection (CSMA/CD) as used on legacy bus-topology Ethernet.
- Carrier sense multiple access with collision avoidance (CSMA/CA) as used on Wireless LANs.

Controlled access

- Deterministic access where each node has its own time on the medium.
- Used on legacy networks such as Token Ring and ARCNET.

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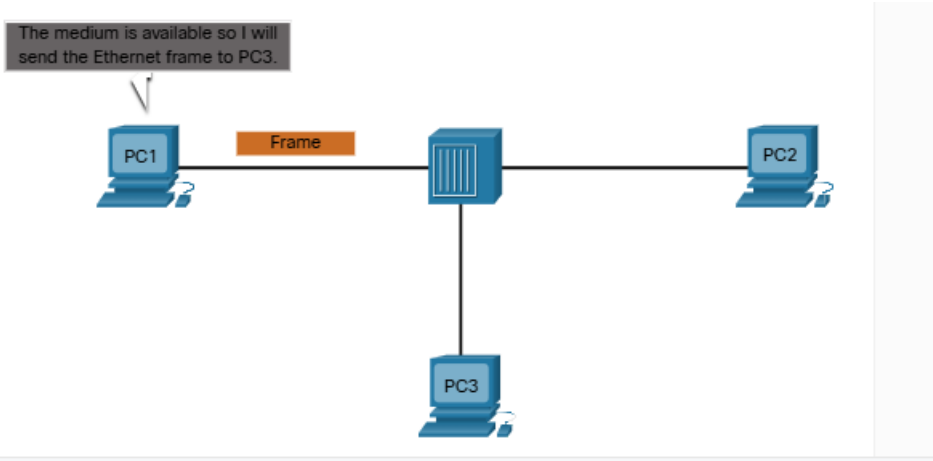
Contention-Based Access – CSMA/CD

CSMA/CD

- Used by legacy Ethernet LANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision detection process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CD collision detection process:

- Devices transmitting simultaneously will result in a signal collision on the shared media.
- Devices detect the collision.
- Devices wait a random period of time and retransmit data.



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Contention-Based Access – CSMA/CD in Ethernet

- Ethernet adapters perform CSMA/CD by measuring ***voltage levels*** before and during transmission
- Each ethernet NIC runs CSMA/CD protocol without coordination with other NICs on the network.
- Within each NIC, the CSMA/CD protocol works as follows:

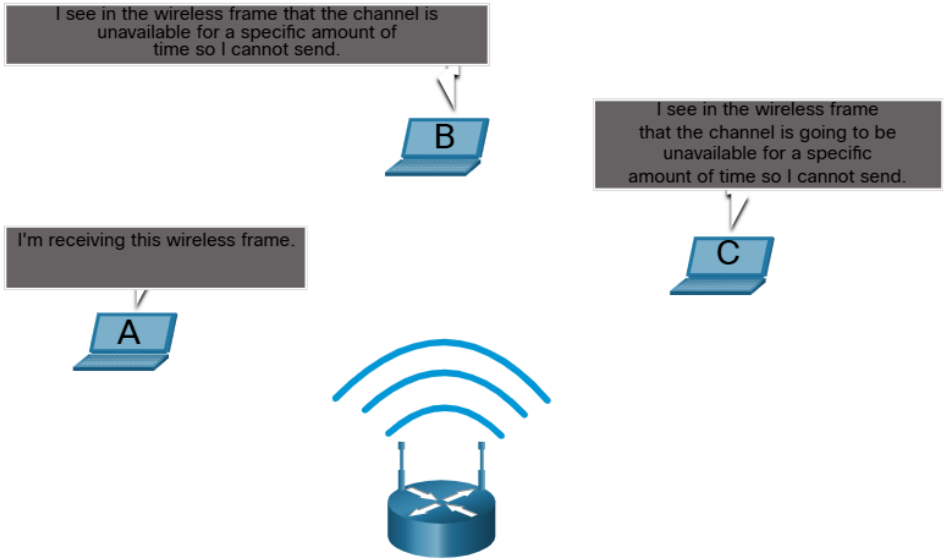
1. The NIC obtains a packet from the network layer, prepares an Ethernet frame, and puts the frame in an adapter buffer.
2. If the adapter senses that the channel is idle (i.e. not signal energy/voltage) entering the adapter from the channel for 96 bit frames, it starts to transmit the frame. If the NIC senses that channel is busy, it waits until it senses no voltage for 96 bit times and then starts to transmit the frame
3. When transmitting, the NIC monitors for the presence of voltage coming from other NICs. If it transmits the entire frame without detecting signal energy from other adapters, the adapter is finished with the frame.
4. If the NIC detects voltage from other NICs while transmitting, it stops its transmission and instead transmits

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Contention-Based Access – CSMA/CA

CSMA/CA

- Used by IEEE 802.11 WLANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision avoidance process to govern when a device can send and what happens if multiple devices send at the same time.



CSMA/CA collision avoidance process:

- When transmitting, devices also include the time duration needed for the transmission.
- Other devices on the shared medium receive the time duration information and know how long the medium will be unavailable.

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6.3 Data Link Frame

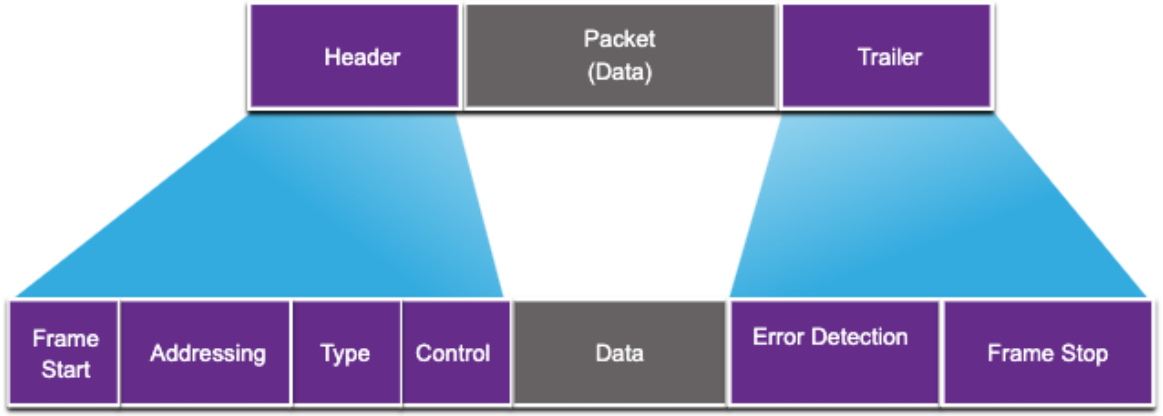
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The Frame

- Data is encapsulated by the data link layer with a header and a trailer to form a frame.
- A data link frame has three parts:
  - **Header**
  - **Data**
  - **Trailer**
- The fields of the header and trailer vary according to data link layer protocol.
- The amount of control information carried within the frame varies according to access control information and logical topology.

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Frame Fields



Field	Description
Frame Start and Stop	Identifies beginning and end of frame
Addressing	Indicates source and destination nodes
Type	Identifies encapsulated Layer 3 protocol
Control	Identifies flow control services
Data	Contains the frame payload
Error Detection	Used for determine transmission errors

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Ethernet Frame Structure

Preamble	Destination address	Source address	Type	Data	CRC

Field	Description	Size (bytes)
Preamble	Identifies beginning of frame	8 bytes
Destination address	MAC address of the destination node	6 bytes
Source address	MAC address of the source node	6 bytes
Type	Identifies encapsulated Layer 3 protocol (i.e. IPV4 or IPV6)	2 bytes
Control	Identifies flow control services	
Data	Contains the frame payload. IPV4 or IPV6 packet	46 to 1500 bytes (MTU – Maximum Transmission Unit). Any packet longer than 1500 bytes is fragmented
CRC	Used for determine transmission errors	4 bytes

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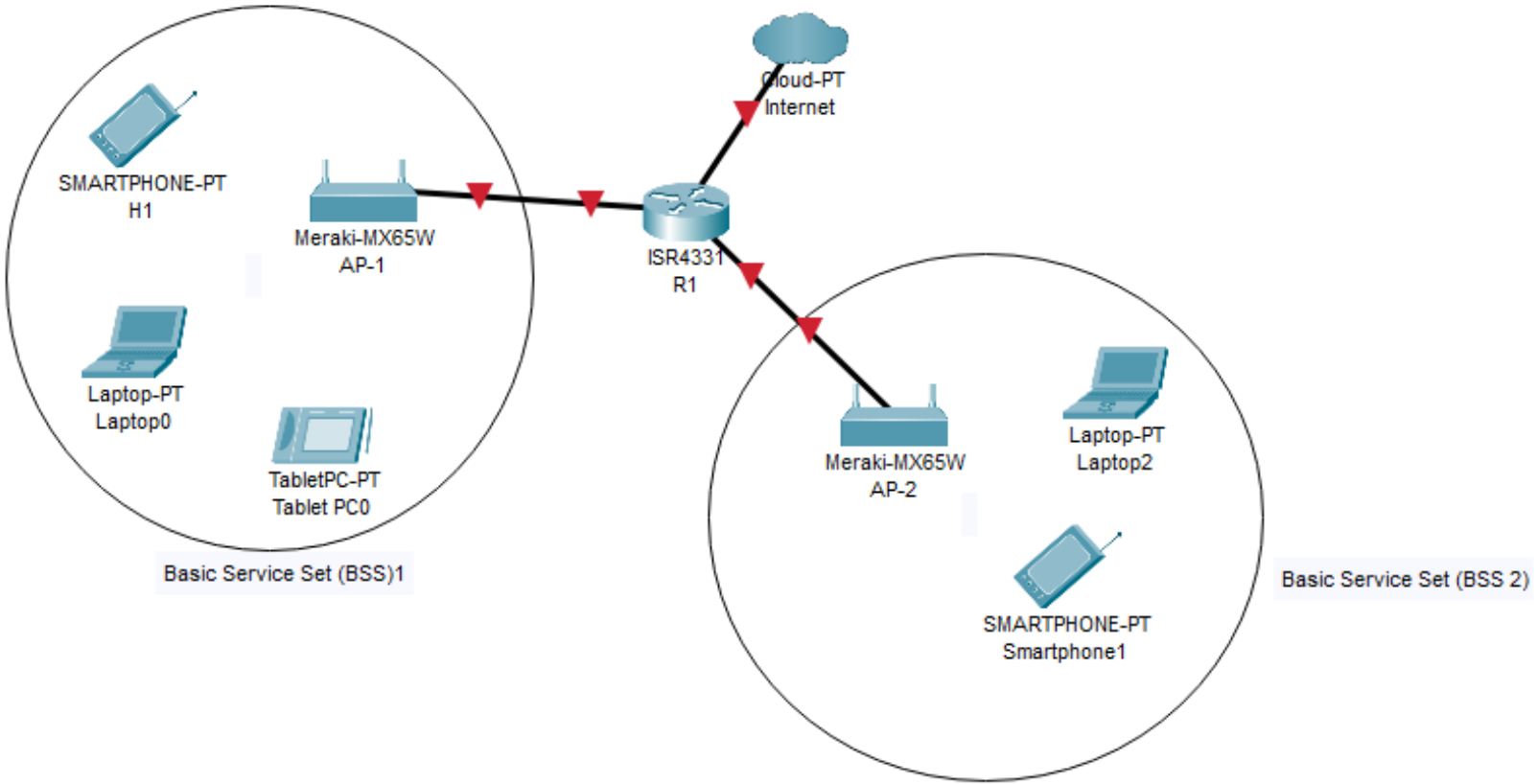
IEEE 802.11 Standard

- IEEE 802.11 wireless LAN, a.k.a WiFi, is a standard for wireless LANs
- All 802.11 standards share most characteristics
- They all use the same medium access protocol (CSMA/CA)
- They all use the same frame structure for their data link layer (L2) frames
- They all allow for both "**infrastructure mode** and **ad hoc mode**"

Standard	Frequency Range	Data Rate
802.11b	2.4-2.485 GHz	Up to 11 Mbps
802.11a	5.1-5.8 GHz	Up to 54 Mbps
802.11g	2.4 GHz	Up to 54 Mbps
802.11n	5 GHz	Up to 600 Mbps
802.11ac	5GHz	Up to 6.9 Gbps
802.11ax	2.4, 5 & 6 GHz	Up to 9.6 Gbps
802.11be	2.4, 5 & 6 GHz	Up to 46 Gbps

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The Architecture of IEEE 802.11



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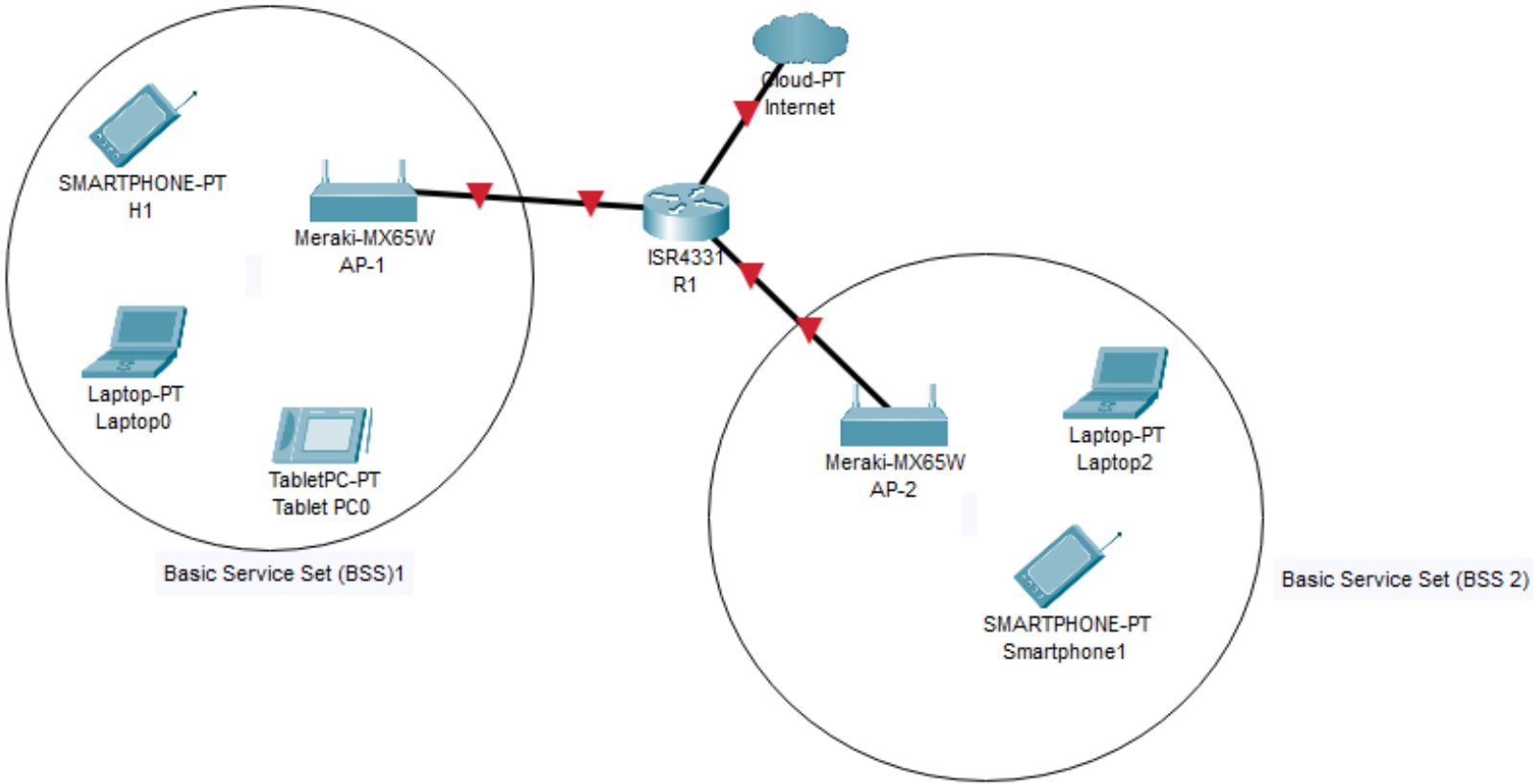
Data Link Frame

IEEE 802.11 Frame Structure

Frame control	Duration	Address 1	Address 2	Address 3	Seq control	Payload	CRC

Field	Description	Size (bytes)
Frame control	Identifies beginning of frame	2 bytes
Duration	Max. time to reserve the channel	2 bytes
Address 1	MAC address of the destination node	6 bytes
Address 2	MAC address of the source node	6 bytes
Address 3	MAC address of the router interface connecting the BSS to the internet	6 bytes
Seq control	Sequence numbers for the receiver to distinguish between new and previous frame	2 byte
Payload	Contains the frame payload. IPV4 or IPV6 packet	0 to 2304 bytes (MTU)
CRC	Used for determine transmission errors	4 bytes

The Architecture of IEEE 802.11



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Frame control

Protocol version	Type	Subtype	To AP	From AP	More frag	Retry	Power mgt	More data	Protected frame	Order

Sub field	Description	Size (bits)
Protocol version	• Identifies IEEE 802.11 protocol version (a,b,c, n, etc.) • Set to 00	2 bits
Type	Indicates frame type • 00 – Management • 01 – Control • 10 - Data	2 bits
Subtype	Distinguishes between request to send (RTS), clear to send (CTS), acknowledgements (ACK), and data frames	4 bits
To AP	• Indicate the direction of the frame • 1 for leaving data, 0 for entering data	1 bit
From AP	• Indicate the direction of the frame • 1 for entering data, 0 for leaving data	1 bit
More frag	Indicates if more fragments are to be received or not	1 bit
Retry	Indicates if the frame is a retransmission of a previously lost frame (1) or not (0)	1 bit
Power mgt	Indicates if the sender will enter power-save mode after transmission (source) (1) or not (0)	1 bit
More data	Indicates if more data is to be received (1) or not (0) (e.g. AP has buffered frames)	1 bit
Protected frame	Indicates whether the payload is encrypted (1) or not (0) e.g. WEP, WPA2, WPA3	1 bit
Order	Indicates whether delivery is ordered (1) or not (0)	1 bit

New Terms and Commands

- Logical Link Control (LLC)
- Medial Access Control (MAC)
- Institute of Electrical and Electronic Engineers (IEEE)
- International Telecommunications Union (ITU)
- International Organization for Standardization (ISO)
- American National Standards Institute (ANSI)
- Physical Topology
- Logical Topology
- Half-duplex
- Full-duplex
- CSMA/CD
- CSMA/CA
- Cyclic Redundancy Check (CRC)
- Contention-based access
- Controlled access

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